Code No. 8407

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. ( AI&ML ) IV-Semester (MAIN) Examinations July-2025

## ARTIFICIAL INTELLIGENCE

100523729008

**Time : 3 hours** **Max. Marks : 60**

**Note :** *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
*ii) Answers must be written in same order as they occur in the Question Paper.*
*iii) **Missing data, if any may suitably be assumed.***

| | | Marks | BT | CO |
|---|---|---|---|---|
| 1. a) | Define Agent and name types of agents. | 2 | 1 | 2 |
| b) | What is state search space? | 2 | 1 | 1 |
| c) | List a few shell tools for developing expert systems. | 2 | 1 | 1 |
| d) | Give how uncertainty is handled in AI. | 2 | 1 | 2 |
| e) | Define heuristic function. | 2 | 1 | 1 |
| f) | Distinguish between crisp and fuzzy sets. | 2 | 4 | 2 |
| 2. a) | Explain in detail about hill climbing search algorithm and also discuss the issues with it. | 6 | 2 | 1 |
| b) | Discuss alpha beta pruning algorithm with an example. | 6 | 6 | 2 |
| 3. a) | Explain various search strategies of problem solving. | 6 | 2 | 1 |
| b) | Illustrate A* algorithm with an example. | 6 | 2 | 2 |
| 4. a) | Explain the architecture of expert systems with the help of neat diagram. | 6 | 2 | 2 |
| b) | Discuss knowledge representation using frames. | 6 | 6 | 1 |
| 5. a) | What is single layer feed forward networks? Explain briefly. | 6 | 2 | 2 |
| b) | Distinguish between expert systems and traditional systems. | 6 | 4 | 2 |

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|---|---|---|---|---|
| 6. a) | Give some of the applications of neural networks. Explain. | 6 | 2 | 2 |
| b) | Explain in detail about Bayesian network. | 6 | 2 | 2 |
| 7. a) | Write short notes on Hopfield networks. | 6 | 2 | 2 |
| b) | Explain about minimax procedure in games. | 6 | 2 | 1 |

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